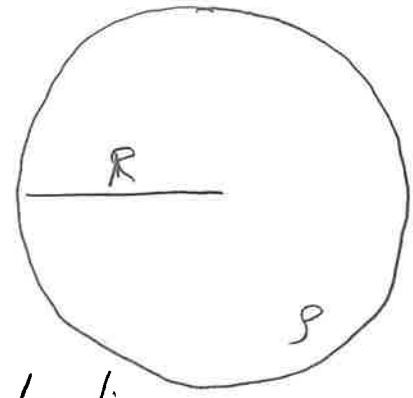
ASTR 430 HW #1 Solutions

(a) Problem #17 from "Building Intuition" Handout
– Gravitational freefall time



units $(t) = s$

units $(\rho) = kg/m^3$

units $(R) = m$

units $\left(\frac{GM}{R^2}\right) = m/s^2 \longleftarrow$ acceleration

$\Rightarrow$ units $(GM) = m^3/s^2$ units $(G\rho) = \frac{1}{s^2}$

units $(G) = m^3/kg\,s^2$

so ...

✗ A) units $\left(\sqrt{\frac{1}{G\rho^2}}\right) = \sqrt{\frac{kg\,s^2}{m^3} \cdot \frac{m^3}{kg}} \neq s$

✗ B) units $\left(\sqrt{\frac{R}{G\rho}}\right) = \sqrt{\frac{m\,kg\,s^2}{m^3}\,\frac{m^3}{kg}} = s\sqrt{m} \neq s$

✗ C) units $\left(R\sqrt{\frac{1}{G\rho}}\right) = \sqrt{m} * B) = s\,m \neq s$

✓ D) units $\left(\sqrt{\frac{1}{G\rho}}\right) = \frac{1}{m} * C) = s$ $\boxed{\text{D wants}}$

✗ E) units $\left(\sqrt{\frac{1}{RG\rho}}\right) = \sqrt{\frac{1}{m}} * D) = \frac{s}{\sqrt{m}} \neq s$

Note the remarkable point that the collapse time is <u>independent</u> of the cloud's radius! A bigger cloud has further to collapse, but more mass to make it collapse faster. <u>Dimensional Analysis</u> can be used to <u>prove</u> this.

Try it! :)

Finally, although D) has the correct units, it is <u>not</u> guaranteed to be right! Units cannot help with the numerical factor out front.

1b) Problem #36 from "Building Physical Intuition in Mechanics"

→ Area of a regular polygon of n sides w/ length x?

A) $A = nx^2\left[\cos(\pi/n)/\sin(\pi/n)\right]$

B) $A = \frac{1}{4}nx^2\left[\cos(\pi/n)/\sin(\pi/n)\right]$

C) $A = nx^2\left[\cos(n\pi)/\sin(n\pi)\right]$

D) $A = \frac{1}{4}nx^2\left[\sin(\pi/n)/\cos(\pi/n)\right]$

All units check out correctly ✓

Try square w/ $n=4$ ⇒ should get x^2

for $n=4$: Expect x^2

A) $A = 4x^2\,[1]$ X

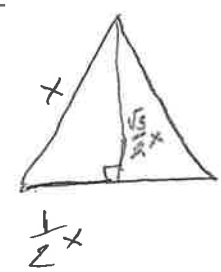
B) $A = x^2\,[1]$ OK

C) $A \to \infty$ X

D) $A = x^2$ OK

$n=3$

Expect $A = \frac{1}{2}x \cdot \frac{\sqrt{3}}{2}x = \frac{\sqrt{3}}{4}x^2$



A) $A = 3x^2\left[\,1/2 \,/\, \sqrt{3}/2\,\right] = \sqrt{3}x^2$ X

B) $A = \frac{3}{4}x^2\left[\,1/2 \,/\, \sqrt{3}/2\,\right] = \frac{\sqrt{3}x^2}{4}$ OK

C) $A = 3x^2\left[\,-1/0\,\right] \to \infty$ X

D) $A = 3/4\,x^2\left[\,\sqrt{3}/2 \,/\, 1/2\,\right] = \frac{3\sqrt{3}}{4}x^2$ X

Try $n=1$:

This is dangerous, but a line segment of length 0 might connect a point to itself. Expect $A=0$, but this edge case may fail.

A) $A=0\,[\infty]$ unclear

B) & C) have the same issue

D) $A=0$ ok

$n=2$

Expect $A=0$

here x can be non zero.

A) $A=0$ ok

B) $A=0$ ok

C) $A \to \infty$ X

D) $A=0$ ok

Try $n \to \infty$ for fixed Area, $x \to 0$ such that circumference $nx = 2\pi r$ is finite

Expect $A = \pi r^2 = \pi\left(\frac{nx}{2\pi}\right)^2 = \frac{1}{4\pi} n^2 x^2$

A) $A = nx^2\left[\frac{n}{\pi}\right] = \frac{x^2 n^2}{\pi}$ ✗

$\sin\varepsilon \approx \varepsilon$
$\cos\varepsilon \approx 1$
for small ε

B) $A = \frac{1}{4}nx^2\left[\frac{n}{\pi}\right] = \frac{1}{4\pi}n^2x^2$ OK

C) $A = nx^2$ [undefined] unclear

D) $A = \frac{1}{4}nx^2[0] = 0$ ✗

Summarize!

	$n=1$	$n=2$	$n=3$	$n=4$	$n\to\infty$
A)	?	OK	✗	✗	✗
B)	?	OK	OK	OK	OK
C)	?	✗	✗	✗	?
D)	OK	OK	✗	OK	✗

Note that some tests are inconclusive and that only B) doesn't fail.

1c) Problem # 49

max height of stirred coffee

A) $\Delta h \propto w$

B) $\Delta h \propto w^2$

C) $\Delta h \propto w^3$

Expect $\Delta h = 0$ for $w = 0$

$\Rightarrow$ all answers pass this test

Expect $\Delta h \uparrow$ as $w \uparrow$

$\Rightarrow$ again, all answers pass

Expect same height for w and $-w$

$\Rightarrow$ A), C) and all odd powers of w fail

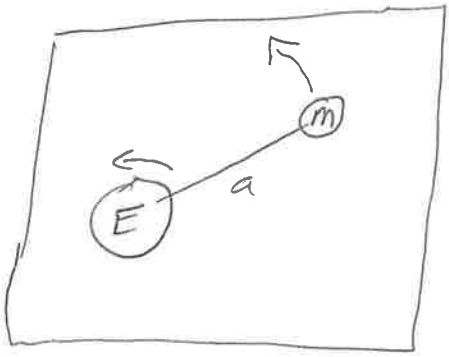
B) still passes ✓

Note, we cannot easily rule out $\Delta h \propto w^4$

1d) Problem #72

Tides raised by the Moon on the Earth push the Moon away from Earth by a few cm per year. What does this do to Earth's rotation rate?

Angular momentum in the box is conserved. (no external torques or mass loss).



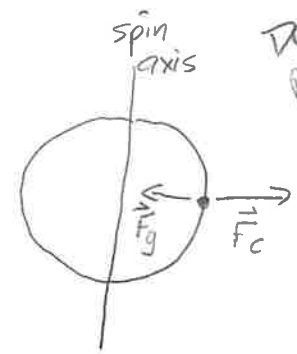
$$L_{TOTAL} \approx m_m \sqrt{G M_E a} + I \omega_{spin}$$

If a increases, ω_{spin} must decrease.

$\Rightarrow$ Earth's spin slows down

2 a) For a spinning star, gravity pulls radially inward while the centrifugal force pushes out.

$$F_g = \frac{GM_*}{r_*^2} \qquad F_c = \frac{v^2}{r_*}$$



Draw a picture!

Equating, we find

$$\frac{GM_*}{r_*^2} = \frac{v^2}{r_*} \Rightarrow v^2 = \frac{GM_*}{r_*}$$

this is just the Kepler speed along a circular orbit. If the star rotates more slowly, it is stable. If it rotates faster, some of its material will be launched into orbit!

The rotation period is then

$$T_{crit} = \frac{2\pi r_*}{v} = \frac{2\pi r_*}{\sqrt{\frac{GM_*}{r_*}}} = 2\pi\sqrt{\frac{r_*^3}{GM_*}}$$

Average Density!

Using $M_* = \frac{4}{3}\pi r_*^3 \rho_*$

$$T_{crit} = \sqrt{\frac{3\pi}{G\rho}}$$

for the Sun, $\rho \sim 1.4\ g/cm^3$

so $T_{crit} = 10{,}000$ seconds

$\boxed{T_{\perp} = 2.8 \text{ hours}}$

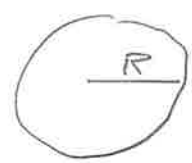
Since T_{crit} depends only weakly on the density of an object, the critical rotation period will not be too different for planets. In fact, for the least dense (Saturn) and most dense (Mercury) planets we have:

Saturn $\rho = 0.69\ g/cm^3 \Rightarrow T_{crit} = 4.0$ hours

Mercury $\rho = 5.43\ g/cm^3 \Rightarrow T_{crit} = 1.4$ hours

A small asteroid, like a basketball, has internal strength which helps to hold it together at higher rotation rates!

b) Moment of inertia of a uniform sphere is $I = \frac{2}{5} M R^2$



Spin Angular Momentum $= I\omega$

Orbital Angular Momentum $= m\sqrt{GM_\odot a(1-e^2)}$

$$\omega_{sun} \approx \frac{2\pi}{26\ \text{days}} \approx 2.8 \times 10^{-6}\ s^{-1}$$

$$R = 696{,}000\ km, \quad M_\odot = 2 \times 10^{30}\ kg$$

So $L_{sun} = \frac{2}{5}(2*10^{30}\,kg)(6.96*10^{8}\,m)^2(2.8*10^{-6}\,s^{-1})$

$= 1.1*10^{42}\,\frac{kg\,m^2}{s}$

$L_{Jup,spin} = \frac{2}{5}(1.9*10^{27}\,kg)(7.1492*10^{7}\,m)^2\left(\frac{2\pi}{9.84\,hours}\right)$

$= 6.9*10^{38}\,\frac{kg\,m^2}{s} = 6.3*10^{-4}\,L_{sun}$

$L_{Jup,orbit} = (1.9*10^{27}\,kg)\sqrt{(6.67*10^{-11})(2.0*10^{30}\,kg)\left[5.2*1.49*10^{11}\,m\right]}$

$= 1.93*10^{43}\,\frac{kg\,m^2}{s} = 17.5\,L_{sun}$

Note $L_{Jup,orbit} >> L_{sun} >> L_{Jup,spin}$

c) $L_{TOTAL} = 2.04*10^{43}\,\frac{kg\,m^2}{s}$

$L_{TOTAL} = \frac{2}{5}M_{\odot}R^2\omega \Rightarrow \omega = \frac{L_{TOTAL}}{\frac{2}{5}M_{\odot}R^2}$

$\omega = 5.2*10^{-5}\,s^{-1}$

so $T = \frac{2\pi}{\omega} = 1.4$ days

Since $T > T_{crit}$, the Sun would not break up, although it would have a larger equatorial bulge. For the real Sun, which is centrally concentrated, the moment of Inertia is $0.059\,MR^2$, 6.8 times less than a uniform sphere.

Since $\frac{I_{(real\ Sun)}}{I_{(uniform\ Sun)}} = \frac{L_{(real)}}{L_{(uniform)}} = \frac{T_{(uniform)}}{T_{(real)}}$

the real Sun would spin 6.8 times faster than we've calculated, $T_{(real)} = 5$ hours. This is quite a bit closer to T_{crit}. If we added in the other planets, the Sun would rotate still faster.

d) $\frac{2}{5} m_\odot R^2 \omega = L_{TOTAL}$

$R = 1 \text{ light year} = 3 \times 10^8 \, m/s \cdot (3600 \cdot 24 \cdot 365) s$

seconds per year

$= 9.5 \times 10^{15} \, m$

$\frac{2}{5}(2 \times 10^{30} \, kg)(9.5 \times 10^{15} \, m)^2 \, \omega = 2.04 \times 10^{43} \, \frac{kg m^2}{s}$

$\omega = 2.8 \times 10^{-19} \, s$

so $T = \frac{2\pi}{\omega} = 2.2 \times 10^{19} \, s$

$= 700$ billion years

This is <u>very</u> slow! The Sun goes around the galactic center in about 200 million years. We expect that the cloud that formed the Sun was larger and rotating faster than

we've calculated here. The early solar system "got rid of" excess angular momentum via strong solar winds; this idea is consistant with the strong stellar winds observed around all young stars today.

3. Structure of Exosphere

$$\frac{dP}{dz} = -\rho g \qquad PV = NkT$$

$$P = \frac{NkT}{V}$$

V = Volume

and $\rho = \frac{M}{V}$ $\quad$ M = mass of atmosphere in V

and $\frac{N}{M} = \frac{1}{m'}$ $\quad$ m' = average molecular mass

$$\Rightarrow \quad P = \frac{kT\rho}{m'}$$

$$\frac{d\rho}{dz} = \frac{-m'g}{kT}\rho$$

$\leftarrow$ we derived this in class.

but now $g \neq$ constant

$$g = \frac{GM_E}{(R_E+z)^2}$$

R_E = Earth's Radius
M_E = Earth's Mass

$$\frac{d\rho}{\rho} = \frac{-m'GM_E}{kT}\frac{dz}{(R_E+z)^2}$$

$$\ln \rho = \frac{-m' G M_E}{kT} \int \frac{dz}{(R_E+z)^2}$$

let $u = R_E + z$

$du = dz$

$$\Rightarrow \int = \int \frac{du}{u^2} = -\frac{1}{u}$$

$$= -\frac{m' G M_E}{kT}\left(-\frac{1}{R_E+z}\right) + K$$

$$= \frac{m'}{kT} \cdot \left(\frac{GM_E}{R_E^2}\right) \cdot R_E^2 \cdot \frac{1}{R_E+z} + K$$

$$= \frac{R_E^2}{H_0} \frac{1}{R_E+z} + K$$

$$\boxed{g_0 = \frac{GM_E}{R_E^2} \qquad H_0 = \frac{kT}{m' g_0}}$$

$$\rho = e^K e^{\frac{R_E^2}{H_0(R_E+z)}}$$

$$\rho(z=0) = \rho_0 = e^K e^{R_E/H_0}$$

$$\Rightarrow e^K = \rho_0 e^{-R_E/H_0}$$

$$\rho = \rho_0 e^{-\frac{R_E}{H_0} + \frac{R_E^2}{H_0}\left(\frac{1}{R_E+z}\right)}$$

$$\rho = \rho_0 e^{\left(\frac{R_E}{R_E+z} - 1\right)\frac{R_E}{H_0}}$$

$$\rho = \rho_0 e^{-\frac{z}{R_E+z}\frac{R_E}{H_0}}$$

Is $T \approx$ const. OK? $T \approx 1000$ K in the upper part of Fig 5.1a. It varies day to night and with solar activity. Probably by a factor of ≈ 2. OK ✓

Units OK ✓

$$\boxed{\rho = \rho_0 e^{-\frac{z}{H'}} \quad \text{w/} \quad H' = H_0 \frac{R_E+z}{R_E}}$$

Now check limits and interpret the answer.

✓ $z=0$: $\rho = \rho_0$ (by definition)

✓ small z : $\rho = \rho_0 e^{-z/H_0}$ (constant gravity case)

✓ $z = R_E$: scale height here should be $4H_0$ since gravity is $4\times$ weaker than at Earth's surface.

$$\rho(z=R_E) = \rho_0 e^{-\frac{R_E}{2H_0}}$$

$$\rho(z=R_E+\delta) = \rho_0 e^{-\frac{R_E+\delta}{2R_E+\delta}\frac{R_E}{H_0}} \leftarrow \boxed{\text{Taylor expand for } \delta \ll R_E}$$

$$\approx \rho_0 e^{-\frac{R_E}{2H_0}\left(1+\frac{\delta}{R_E}\right)\left(1-\frac{\delta}{2R_E}\right)}$$

(δ is height above $z=R_E$)

$$\approx \rho_0 e^{-R_E/2H_0\left(1+\frac{\delta}{2R_E}\right)}$$

$$= \rho(z=R_E)\, e^{-\delta/4H_0}$$

(scale height is $4\times$ larger as expected)

✓ $z \to \infty$: $\rho = \rho_0 e^{-R_E/H_0} =$ constant!

why? far from Earth there is no gravity and in equilibrium ρ and P are (very small) constant.